

AUDITORY GROUPING DAILY

Saturday, 11 July 2026 · 369 papers screened · curated

Today's deterministic feeds returned almost nothing in your core paradigm — of 364 automatically-gathered candidates, essentially none touched chunking, auditory electrophysiology, or sequence-based decision-making. A supplementary watch-author search filled the gap and turned up a real cluster: prediction-error architecture across the auditory hierarchy, from inferior-colliculus single units through cortical-layer fMRI to distributed intracranial EEG, plus a large-scale study of age-related, non-stationary neural variability during decision-making. The throughline is that 'prediction error' fractures into sub-components across these papers (adaptation versus genuine prediction error at the midbrain; superficial versus deep cortical layers; overlapping rather than segregated frequency bands), directly relevant to how confidently you can label a signal in your prelimbic or auditory-cortex recordings as 'the' prediction-error response. Honest note: this is a manually-assembled day, not a naturally rich one — treat the small count as a quiet week in the automated feeds, not under-effort.

CHUNKING & SEQUENCE PROCESSING

Impact of Education and Music Training on the Development of Abstract Thinking in the First Years of Schooling

CROSSREF (MODEL-DISCOVERED) Théo Morfousse et al. — *Open Mind* — 2026

Corresponding: Stanislas Dehaene — Collège de France — France | [Open paper](#)

Testing 528 children aged 4-8 on how they encode and compress auditory tone sequences and visual patterns, this paper finds that by first grade children already compress sequences using a 'Language of Thought'-style symbolic code similar to adults, with no additional benefit from several years of violin training. Because your human chunk-in-stream task is explicitly probing the transition between unconscious statistical extraction and conscious template-based chunking, this is a useful adjacent data point: it suggests the capacity to compress an auditory sequence into a compact symbolic chunk is already largely in place early in development and isn't obviously trained by musical experience, which bears on how you'd interpret individual differences in chunking ability as reflecting a fixed capacity rather than acquired expertise.

STATISTICAL LEARNING & ODDBALL

Temporal dynamics of contextual processing in the inferior colliculus of a rat model of autism: Sex- and age-dependent trajectories

CROSSREF (MODEL-DISCOVERED) S. Cacciato-Salcedo et al. — *Hearing Research* — 2026-07

Corresponding: M.S. Malmierca — University of Salamanca — Spain | [Open paper](#)

Using oddball and cascading paradigms in the non-lemniscal inferior colliculus, this paper cleanly dissociates three components of context-dependent auditory processing — mismatch response, repetition suppression, and prediction error — and shows they mature on different, sex-dependent trajectories, with prenatal valproic-acid exposure disrupting the pattern. This is useful groundwork for your mouse sequence-discrimination task: it demonstrates that 'prediction error' in the auditory pathway is not monolithic even at the earliest subcortical stage, worth keeping in mind when you interpret prediction-error-like modulation in your prelimbic or auditory cortex recordings as a single mechanism rather than a mixture of adaptation and genuine prediction. The cascade-plus-oddball design is also a clean methodological template if you ever want a repetition-suppression control condition for your target-versus-distractor comparisons.

Predictive acoustical processing in human cortical layers

CROSSREF (MODEL-DISCOVERED) Lonike K. Faes et al. — *Nature Communications* — 2026-05-26

Corresponding: Federico De Martino — Maastricht University — Netherlands | [Open paper](#)

Using laminar-resolution fMRI during a cascading oddball paradigm, this paper localizes prediction-violation signals to superficial layers of the planum polare and middle layers of lateral temporal cortex, while model-updating signals appear in deep layers — a layer-specific dissociation that maps predictive-coding theory onto measurable human cortical architecture. This directly complements today's Paper of the Day and gives you a concrete, testable layer-based hypothesis for your own auditory-cortex recordings: if you ever get depth information from your Neuropixels probes, this predicts that prediction-error-like responses to an unexpected chord should arise more superficially than the slower model-updating signal that follows it.

LANGUAGE, STROKE & BCI

The effects and interactions of top-down influences on word recognition

CROSSREF (MODEL-DISCOVERED) Reuben Chaudhuri et al. — *Hearing Research* — 2026-08

Corresponding: Jan Schnupp — City University of Hong Kong — Hong Kong SAR | [Open paper](#)

In an EEG study of spoken-word recognition in noise, this paper separates two sources of top-down expectation — informative semantic hints versus simple stimulus repetition — and finds they operate on different timescales: hint effects fade quickly across repetition blocks while repetition-based facilitation is stronger and more persistent, dominating when both cues are present together. This maps onto the same statistical-learning-versus-template-matching distinction you use to frame your own work: the finding that repetition-based learning outlasts and overrides hint-based expectation is a concrete precedent for expecting unconscious pattern extraction to dominate over conscious template matching once a sequence has been heard enough times, directly testable across your task's five-stage difficulty progression.

CIRCUITS, METHODS & POPULATION ANALYSIS

Age-related changes in behavioural and neural variability in a decision-making task

CROSSREF (MODEL-DISCOVERED) Fenying Zang et al. — *Nature Communications* — 2026-06-30

Corresponding: Anne E. Urai — Leiden University — Netherlands | [Open paper](#)

Using more than 18,000 single units recorded across 16 brain regions — including striatum, midbrain, and thalamus alongside cortex — via the International Brain Laboratory's standardized visual decision-making task, this paper shows that ageing increases post-stimulus neural variability and reduces 'variability quenching' (the drop in trial-to-trial variability that normally follows stimulus onset), most strongly in visual and motor cortex, striatum, and thalamus. This is a direct empirical instance of treating neural variability itself, not just mean firing rate, as the target of analysis — the same spirit as your GLM's Module C for slow drifts in strategy from satiety or time-on-task. Here the 'drift' is across age rather than within a session, but the analytic move of quantifying how tightly a population's activity is quenched by a stimulus versus left noisy is directly portable to asking whether your prelimbic or auditory-cortex populations show session-long changes in variability quenching as mice gain experience with the target sequence.

PAPER OF THE DAY

Neural mechanisms of time-forward predictions for naturalistic auditory tone sequences.

Thomas J Baumgarten et al. — Nature Communications — 2026 — new pick — abstract only

Corresponding: Biyu J He — New York University Langone Health — USA | [Open paper](#)

- Using intracranial EEG recorded directly from cortex in neurosurgical patients, and stimuli built to have naturalistic statistical regularities rather than the simple repeating patterns most prediction studies use, the authors report that time-forward prediction signals were carried mainly by low-frequency activity spread across sensory, parietal, and frontal cortex, while prediction-error signals showed up in both low- and high-frequency activity, with the high-frequency component largely confined to sensory regions.
- The paper's central claim is a negative result for a common assumption: prediction and prediction-error signals did not show the clean spatial or spectral segregation that earlier hypotheses expected, meaning you can't assume a given frequency band or region is doing only one of those two jobs at once.
- Directed connectivity between regions dropped over the course of a sequence as its predictability increased, except along pathways originating in parietal cortex, which stayed elevated — the abstract frames this as evidence for an integrative, distributed predictive process with a sustained top-down contribution from parietal cortex and the dorsal auditory pathway, rather than prediction being resolved once and then handed off.
- Because the abstract itself doesn't specify exactly what 'naturalistic statistical regularities' means for the tone sequences (only that they're closer to natural auditory statistics than typical lab stimuli), this is the one claim in the paper worth verifying against the full Methods before you lean on it too hard as a direct analogue to your own tone/chord sequences — the abstract's framing is credible but the specific regularity structure isn't described at this level of detail.

HOW TO READ IT (≈20 min)

Full text wasn't available for this pass (Elicit access is also not currently enabled on this account), so treat this as an abstract-only read plan to refine once you have the PDF. Start with the Introduction to see their specific critique of the low-frequency-predicts/high-frequency-errors assumption (2-3 min). Go straight to Methods and read closely exactly how the 'naturalistic' tone-sequence statistics were generated and which electrode coverage regions counted as sensory/parietal/frontal (5-6 min) — this is the part the abstract underspecifies and where your own judgment about relevance to tone/chord sequences will actually be decided. Read the Results on spatial/spectral overlap in full (6-7 min). Read the directed-connectivity results carefully, especially the parietal-cortex exception (4-5 min). Skim the Discussion for how they reconcile the overlap finding with predictive-coding theory (2 min).

YOUR QUESTION TO ANSWER

The abstract reports that parietal-cortex outgoing connectivity stayed high even as overall directed connectivity fell with increasing predictability — if that finding survives a look at the full Methods, would you expect an equivalent 'exception' region in your own mouse prelimbic-versus-auditory-cortex comparison (i.e., a region whose outgoing influence on the other stays elevated even once the mouse has learned the target sequence well), or does your task's much smaller state space make that kind of sustained top-down channel less likely to be needed? And is your current pipeline (single-unit and LFP recordings in prelimbic cortex, later auditory cortex) actually set up to compute directed connectivity between the two regions, or would testing this require adding a specific analysis you haven't run yet?